25900642 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

THIRD SEMESTER

MG3DSESDV201 - INTRODUCTION TO MEDIA FORMATS
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 70**

Part A

Very Short Answer (Answer any 8) 2 Marks Each

1. Can you tell if a file was compressed using lossy or lossless compression just by looking at it?

   No, not typically just by looking at the file itself. You would need to check the file format/codec or compare it to the original.

2. Define "clip description" metadata and its importance in media workflows.

   Metadata that provides descriptive information about a media clip (e.g., scene, take, notes). It's important for organization, searching, and understanding content quickly in a large workflow.

3. How can proxies improve the editing experience for editors with slower computers or systems?

   Proxies are low-resolution, low-bitrate versions of the original media, making them less demanding on the computer's CPU and GPU, thus allowing for smoother, real-time playback and editing.

4. What is RAID 0, and what are its advantages and disadvantages in data storage?

   RAID 0 (Striping): Data is split and written across multiple drives. Advantages: Fastest performance, no capacity loss. Disadvantages: No redundancy (if one drive fails, all data is lost).

5. Why is it important to back up external drives regularly in professional media workflows?

   To prevent data loss due to drive failure, corruption, theft, or human error, ensuring that projects and valuable media assets are safe and recoverable.

6. What is the primary function of a digital camera's sensor?

   To convert light into electronic signals (data) that form the digital image or video.

7. How does the use of bins help in large-scale editing projects?

   Bins (folders within editing software) help to organize media files, sequences, and other assets, making it easier to locate and manage thousands of files in complex projects.

8. How do proxies contribute to workflow efficiency?

   By allowing editors to work quickly with smaller, easier-to-process files, reducing rendering times, and minimizing system slowdowns, which speeds up the editing process.

9. Explain how external drives can be integrated into a data recovery plan.

   They serve as a destination for local backups, providing an accessible and often inexpensive source to restore data if the primary storage system fails or is corrupted.

10. What is the difference between .mov and .mp4 container formats?

    Both are media container formats. .mov (QuickTime File Format) is developed by Apple. .mp4 (MPEG-4 Part 14) is an international standard and is generally more universally compatible for web/playback.

Part B

Short Answer (Answer any 5) 6 Marks Each

11. Describe the potential pitfalls when working with proxy media in editing software and how to avoid them.

    Pitfalls: Forgetting to switch back to original media before export, resolution/frame rate mismatch, or accidental deletion of original media.
    Avoidance: Clearly label/organize files; use a systematic relink/match back process; maintain rigorous backup of original files.

12. Compare different backup strategies for small-scale vs. large-scale productions, considering factors like budget, team size, and media storage requirements.

    Small-Scale: Lower budget and storage; simpler strategy like "Drive 1 + Drive 2" or a single cloud sync.
    Large-Scale: Higher budget and massive storage; requires robust systems like LTO Tape, RAID 5/6, or enterprise-level cloud solutions for redundancy and scalability (3-2-1 Rule).

13. Provide a comprehensive explanation of how to choose the right bitrate for a video project, considering factors like resolution, frame rate, and delivery method.

    Bitrate determines data flow (quality/file size). Higher resolution/frame rate generally needs a higher bitrate to maintain quality.
    The delivery method is key: Streaming (e.g., YouTube) has specific, moderate bitrate recommendations; Broadcast/Archive needs very high bitrates for maximum quality. Balance quality, file size, and target platform requirements.

14. How does Google Drive streamline media management for small teams working on collaborative editing projects?

    It provides a centralized, accessible cloud storage where team members can share, review, and organize media files and project assets. Its synchronization features ensure everyone is working with the latest versions of the media and project files.

15. Discuss how resolution and bitrate influence the quality and performance of proxy files. Provide examples of different proxy formats.

    Resolution and Bitrate are kept low for proxies. Lower values mean lower quality but higher performance (smoother editing).
    Examples: ProRes Proxy, H.264 (low bitrate), Avid DNxHR LB (Low Bandrate).

16. Compare the cost-effectiveness and scalability of cloud backups for large-scale media projects. What are some factors to consider when choosing a cloud storage provider?

    Cost-Effectiveness: High for scalability (only pay for what's used) but can be expensive long-term due to egress fees and cumulative storage costs.
    Scalability: Excellent, offering virtually unlimited storage expansion on demand.
    Factors to Consider: Security/Encryption, Speed (upload/download), Egress Fees, Pricing structure, and Integration with existing workflow.

17. Compare and contrast the process of importing media versus ingesting media in a professional workflow.

    Importing is the simple act of linking a file to an editing project (e.g., linking an MP4).
    Ingesting is a more structured, professional process that often involves copying the files (with verification), backing up, transcoding (converting format), and adding metadata before the files are available for editing.

Part C

Essay (Answer any 2) 12 Marks Each

18. Discuss how to resolve issues related to incompatible proxy formats or settings in an editing project. Provide steps to check and fix common compatibility issues between proxy and full-resolution media.

    Resolution: Ensure the proxy and original media have the same timecode, frame rate, and aspect ratio. Incompatibility often arises from these mismatches or incorrect proxy creation settings.
    Fixing Steps: Check/Verify Original Media Properties (timecode, frame rate). Delete Incompatible Proxies. Re-create Proxies using settings identical to the original media's parameters (except resolution/bitrate). Use the editing software's "Relink/Re-match" function to connect the original media to the new proxies.

19. Explain in detail the importance of implementing redundancy in backup systems and how it helps to prevent data loss in visual editing workflows. Provide examples of redundancy techniques used in RAID systems and mirrored drives.

Importance: Redundancy is having multiple copies of data/system components to ensure continuity even if one fails. It is critical in visual editing where data loss can mean losing weeks of work. Helps prevent loss by allowing a system or workflow to recover instantly from a single point of failure. Examples: RAID 5/6 (uses parity data for drive recovery); Mirrored Drives (RAID 1) (identical copy of data on a second drive).

20. Describe a scenario where poor file organization led to significant delays or complications during post-production. How could this have been prevented through better media management practices?

    Scenario: An editor starts a film by dumping all footage, graphics, audio, and project files into a single, unorganized folder. Months later, a missing sound effect or an old sequence cannot be located, leading to a frantic, time-consuming search that delays the final delivery.
    Prevention: Implement a standardized, consistent folder structure from the start (e.g., separate folders for Footage, Audio, Graphics, Project Files, Exports). Use clear, standardized file naming conventions (e.g., Date_Scene_Take) and use the editing software's Bins (folders) to mirror this structure.